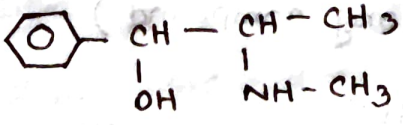
ALKALOIDS

The nitrogenous heterocyclic organic compound found majorly in plants & animal product, usually having basic nitrogen containing complex structure, physiologically active are generally called alkaloids.

General Properties :

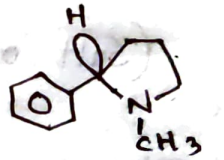
1. Colourless
2. Crystalline
3. Non-volatile
4. Insoluble in H_2O but soluble in other organic solvents
5. Have bitter taste
6. Optically active
7. They are generally tertiary nitrogen in ring system
8. Most of the alkaloids contain oxygen.

Ex :



Ephedrine

,



Nicotine

FARIHA RAHMAN DISHA
CHEM-50

14.03.24

* Extraction or isolation of Alkaloids :

Collected Powder, plant materials

↓ 1. Methanol
↓ 2. Boil
↓ 3. Filter

Filtrate (methanol) —Evaporation of methanol→ Solid Mass

↓

(Alkaloids / methanol soluble organic compound + non-alkaloidal basic compound other neutral compound + acidic organic compound)

5% HCl + 95% H_2O → Salt of basic compounds (Alkaloid/non alkaloid) + water soluble org compound + solid mass

↓ Filter

Solid mass + aq layer

(Salt of basic org compound)

↓ $CHCl_3$

Aq layer
(Salt of basic compound)

$CHCl_3$ layer
(Other water soluble organic compound)

↓ $NaHCO_3$ / Na_2CO_3

Basic Compound (alkaloid + non alkaloid) + Inorganic salt

$CHCl_3$ layer

Aq layer

$CHCl_3$ layer

Free alkaloid + non alkaloid base

① Anhydrous Na_2SO_4 → absorbs water

② Evaporation

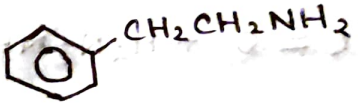
↓

Solid / Semi solid Mass

↓ Chromatographic separation (column)

Individual Free Alkaloids

Classification of Alkaloids :

1. Phenylethylamine group 

2. Pyrrolidine 



3. Pyridine 

4. Pyrrooline - pyridine group
5. Quinoline Group 

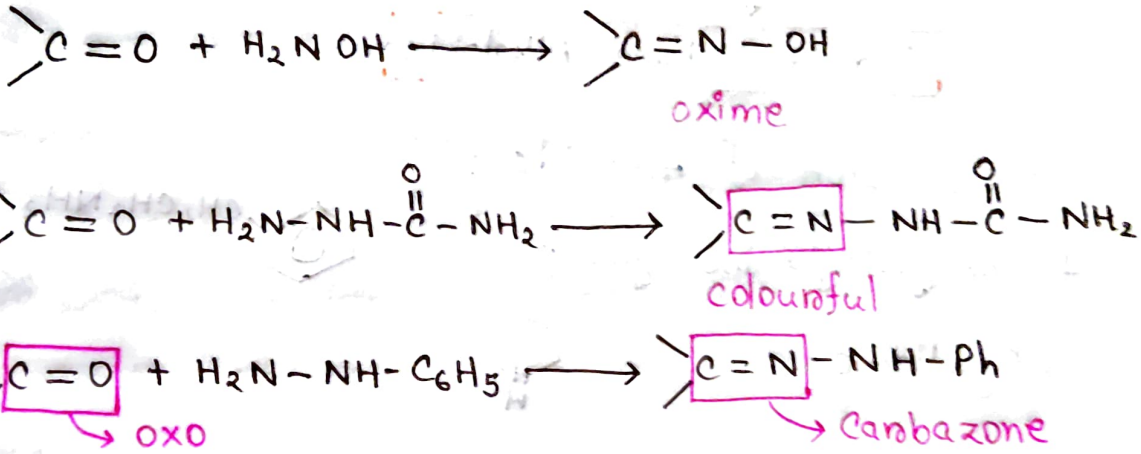
6. Iso-quinoline group
7. Phenanthrene group
8. Indole group

20.03.24

Occurance of Alkaloids :

Alkaloids are chemically heterogeneous group of approximately 2500 basic N containing substance found in about 15% of all vascular plants in more than 150 plant families. They are widely distributed in higher plants particularly the dicotyledons less frequently in fungi.

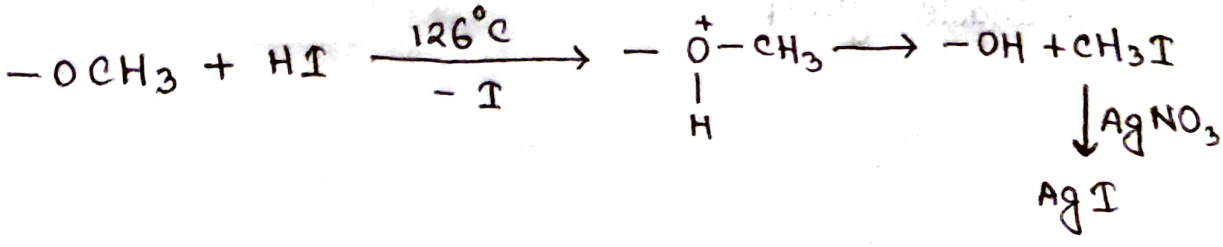
Oxo-group :



Unknown substance এর structure chemically ও spectroscopically determine করা যায়। এর পরেরটা chemical method. $>C=N-$ এর colour দেখে বুঝা যায় oxo-group ছিল।

Methoxy group ($-OCH_3$) :

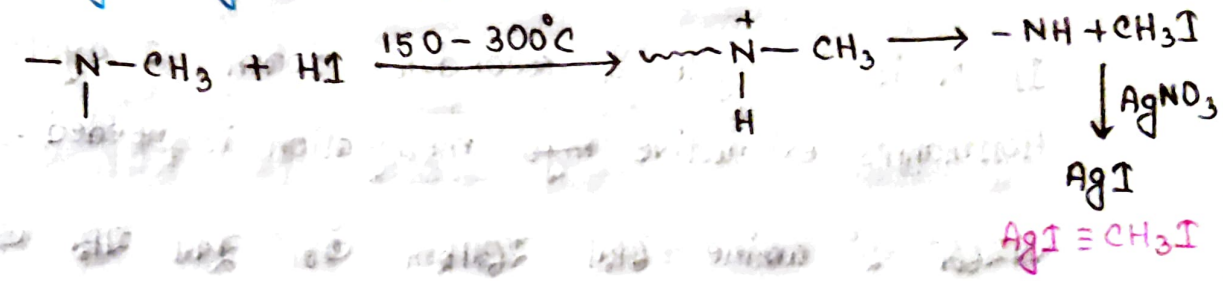
Zeisel Method / determination —



Methoxy group এ HI দিলে প্রথমে protonated হবে। পরে $AgNO_3$ দিলে AgI এর ppt পড়বে, যার সাথে CH_3I এর mole সংখ্যা নির্ণয় করা হয়। CH_3I যদি 2 mole থাকে তবে alkaloid-এ methoxy group-ও 2টি থাকবে।

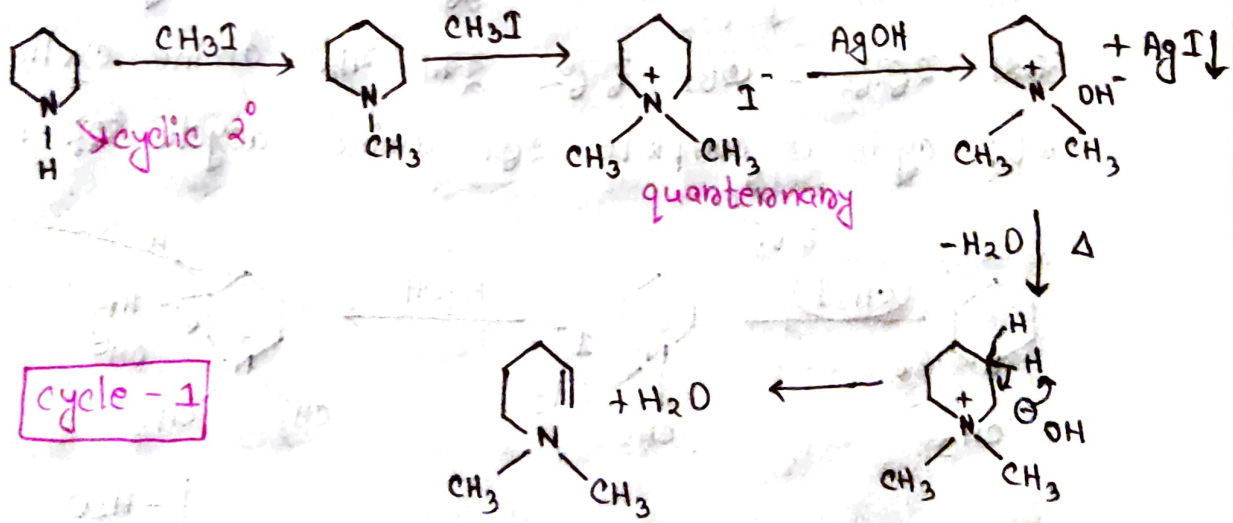
N–CH_3 Group :

Herzig - Meyer Method / Determination



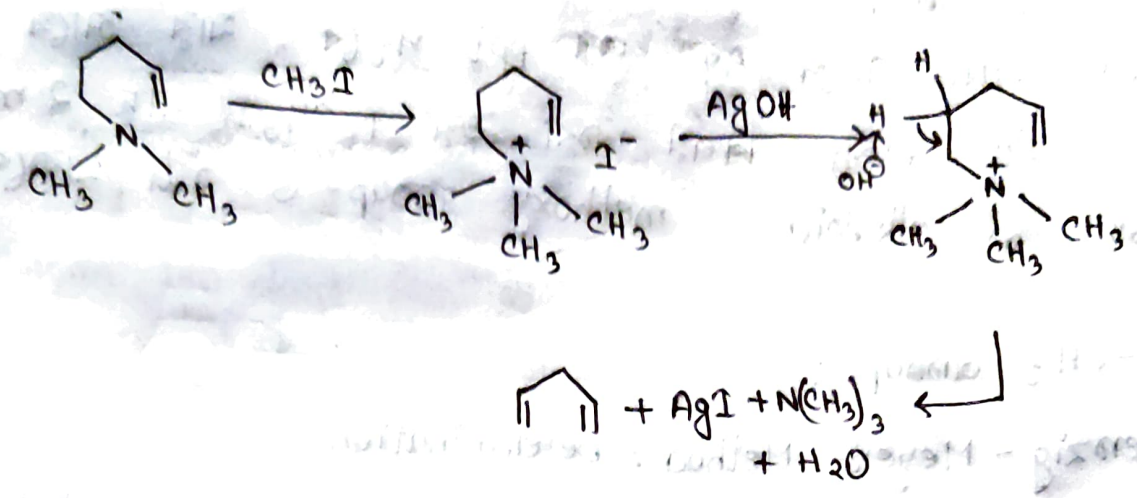
Hoffmann's Exhaustive methylation :

Alkaloid-এ amine group থাকে। এই amine group কত degree তা এই method এ নির্ণয় করে।



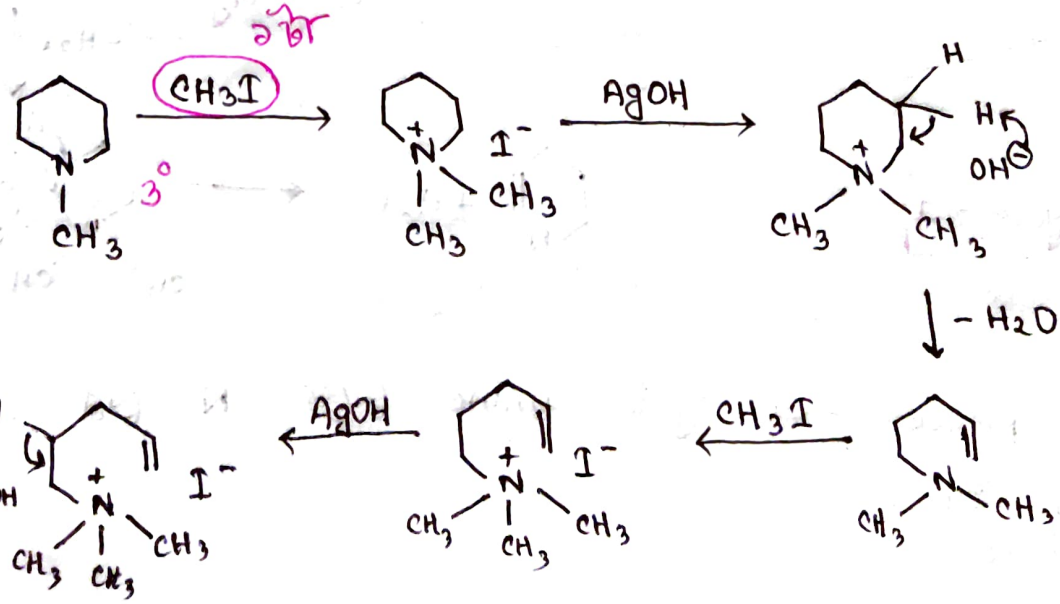
এই process-এ cyclic amine থেকে N বের করে আনা হয়।

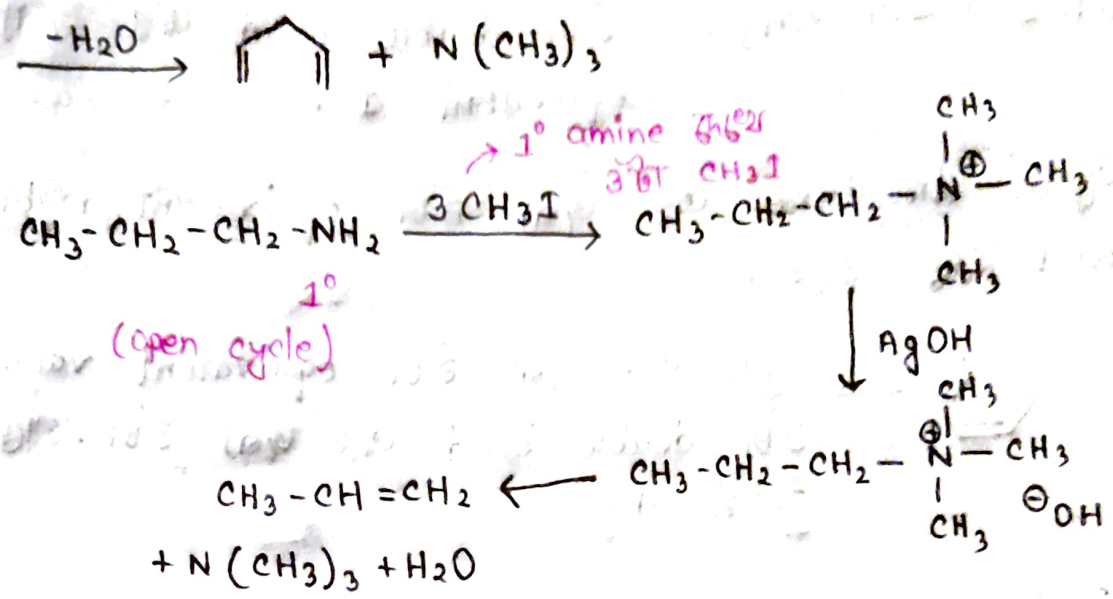
Cycle - 2



If N is in cycle & secondary form, 2 cycle of Hoffmann's exhaustive ~~myt~~ methylation is needed.

প্রথমে 2° amine নেয়া হয়েছিল এটা বুঝা যায় এখানে যে 2 বার CH_3I absorb হয়েছে এর মাধ্যমে। 1° হলে 3টা CH_3I লাগতো। 3° হলে 1টা CH_3I - এই rxn কোষ হত।
আর 2টা cycle এ N - ~~[illegible]~~ completely remove করা হয়েছে - এটা দেখে বুঝা যায় amine cyclic ছিল, 1টা cycle এ complete হলে amine acyclic হয়।





Open cycle ছিল দেখে ৯টা cycle এই rxn complete, মানে N বের হয়ে গিয়েছে।

* Unknown compound এ 1°, 2°, 3° ও cyclic/acyclic amine আছে কিনা কিভাবে determine করা হয়?

* Structure Determination of Ephedrine (Alkaloid) (Chemical Method):

Ephedrine → Anti-asthma drug, stimulant (like caffein), Increasing pressure

Molecular formula → $C_{10}H_{15}NO$ — এই formula থেকে structure guess করতে হবে

↓

প্রথমে alkane এর আকারে আনতে হবে, N থাকলে +1 ও কোনো halogen থাকলে -1 হবে

$C_{10}H_{22+1}$

$C_{10}H_{23}$

$C_{10}H_{15}$

$H_{23} - H_{15} = 8$টা H | ৪ কে ২ দিয়ে ভাগ করতে হবে।

$\frac{8}{2} = 4$ → compound এ ৪টা double bond equivalent থাকবে

$CH_2 = CH - CH_2 - CH_3$ ⟶ ১টা double bond এর জন্য equivalent 1.

Cyclic structure এর জন্য ১টা equivalent double bond ধরতে হবে।

 ⟶ cycle এর জন্য ১টা equivalent value, ৩টা double bond এর জন্য ৩টা, মোট total 4.

যেহেতু আমাদের আগের value-এ 4 আসছে, মানে এতে ১টা benzene ring থাকতে পারে। Ring এ ৬টা C, বাকি আর 4টা C থাকে, যেগুলোতে single bond থাকতে পারে।

এখন, N-এর form বের করতে হবে —

1) Presence of 2° amine

$$C_{10}H_{15}NO \xrightarrow[0-5°C]{HNO_2} C_{10}H_{14}O-N=O$$

Nitroso compound (NH-bond আছে)

HNO_2 এর সাথে 1° amine ⟶ diazonium salt,
2° ⟶ nitroso compound, 3° ⟶ No rxn. যেহেতু nitroso comp,
∴ 2° amine present.

3) Presence of OH:

O এখানে অবশ্যই C=O এ থাকবে না কারণ double bond সব ring এ চলে গিয়েছে।

$$C_{10}H_{15}NO + 2\ Cl-\overset{O}{\overset{\|}{C}}-Ph \longrightarrow 2 \text{ benzoyl derivative}$$

When it is benzoylated, it forms a dibenzyl derivative indicating that O atom is present as hydroxyl & another is $-NH$.

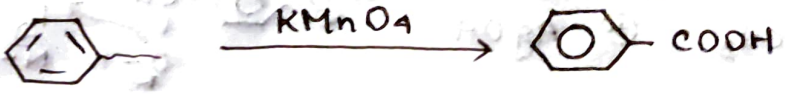
2 mole reactant নেয়া হয়েছে যেখানে 1 mole hydroxyl ও 1 mole $-NH$ এর সাথে rxn করে। O এর সাথে rxn করে benzoyl derivative পাওয়া যায়, যা proove করে hydroxyl group present.

4) hydroxyl এর form নির্ণয় –

$$C_{10}H_{15}NO \xrightarrow{CrO_3} \text{ketone}$$

ketone তৈরি হওয়ায় বলা যায় hydroxyl 2° form এ.

5) When ephedrine is oxidized, it yields benzoic acid, indicating that it contains a benzene ring with a side chain.



Mono-substituted benzene ring present

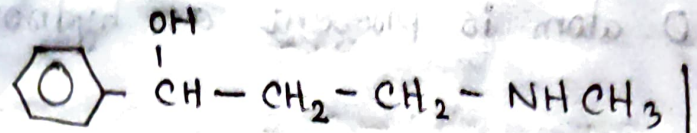
6) Nature of side chain:

When ephedrine is heated conc HCl, it forms methyl amine, which reveals $N-CH_3$ present.

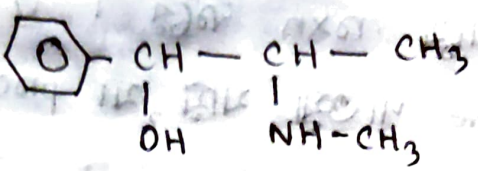
$$C_{10}H_{15}NO \xrightarrow{HCl} CH_3NH_2 + C_6H_{15}CO\,CH_2CH_3$$

24.03.24

Possible structures :



(I)

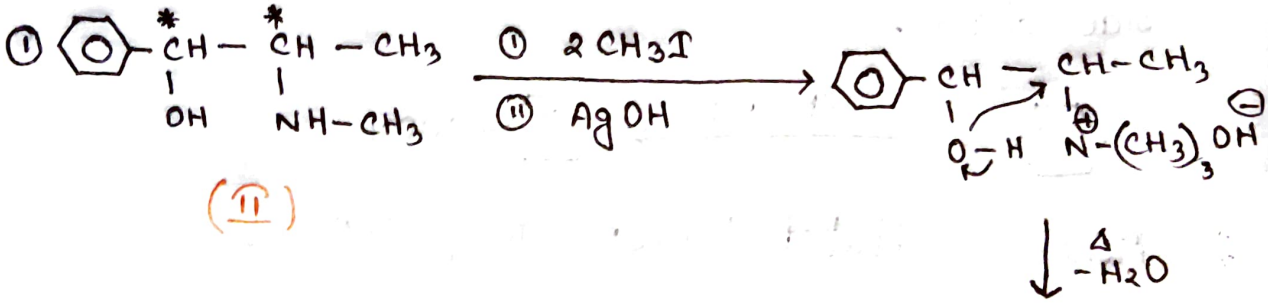


(II)

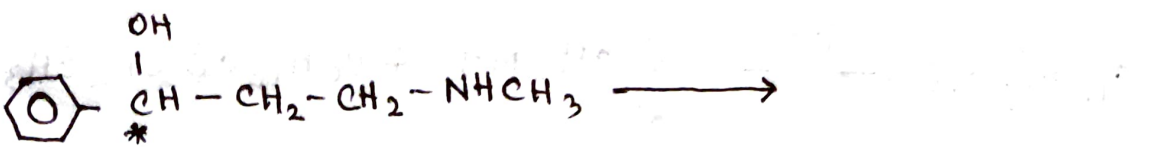
Chemically এই bond গুলো পাওয়া যায়—

- -NH
- $-NH-CH_3$
- $-CH-OH$
- 

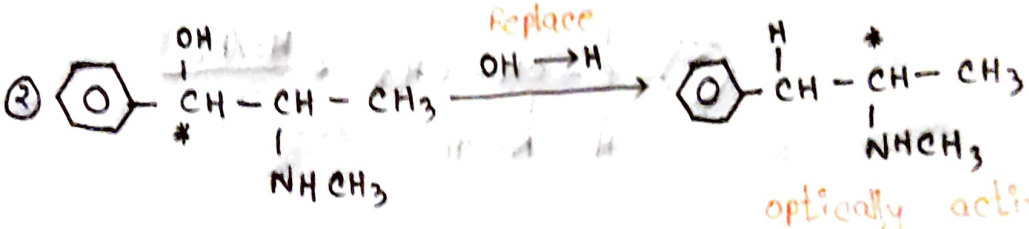
The structure II is supported by the fact that ephedrine when subjected to Hoffmann's exhaustive methylation method, it yields 1,2-methylphenylethoxide which is structure II.

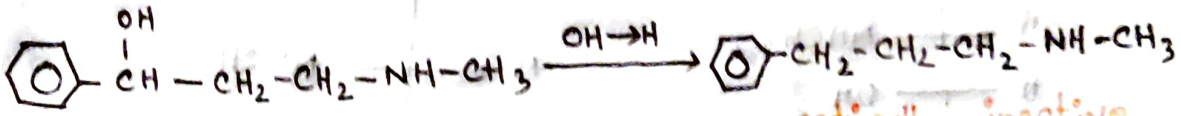


(II)



Ephedrine optically active. Real ephedrine এর sample এ chiral center এর OH কে H দিয়ে replace করলেই যে optical activity পাওয়া যায়, structure ① & ② এও OH replace করলে যেটায় same activity পাওয়া যাবে, সেটাই ephedrine এর structure.





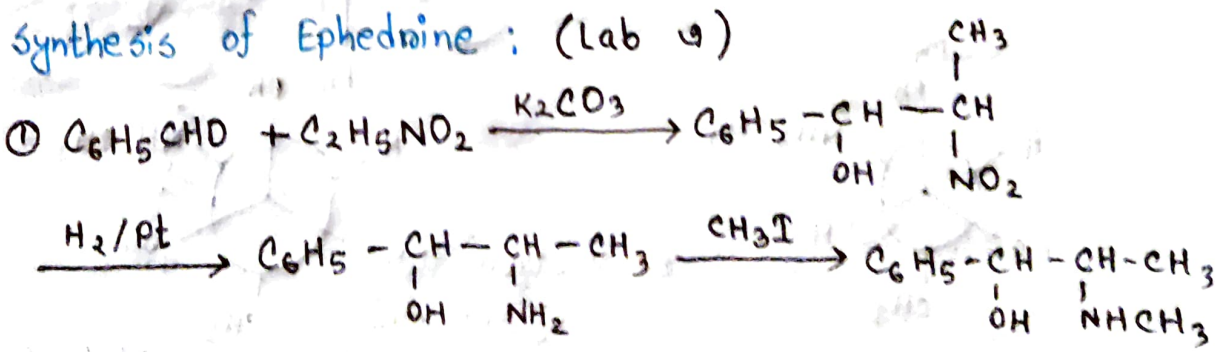
optically inactive
(No chiral carbon)

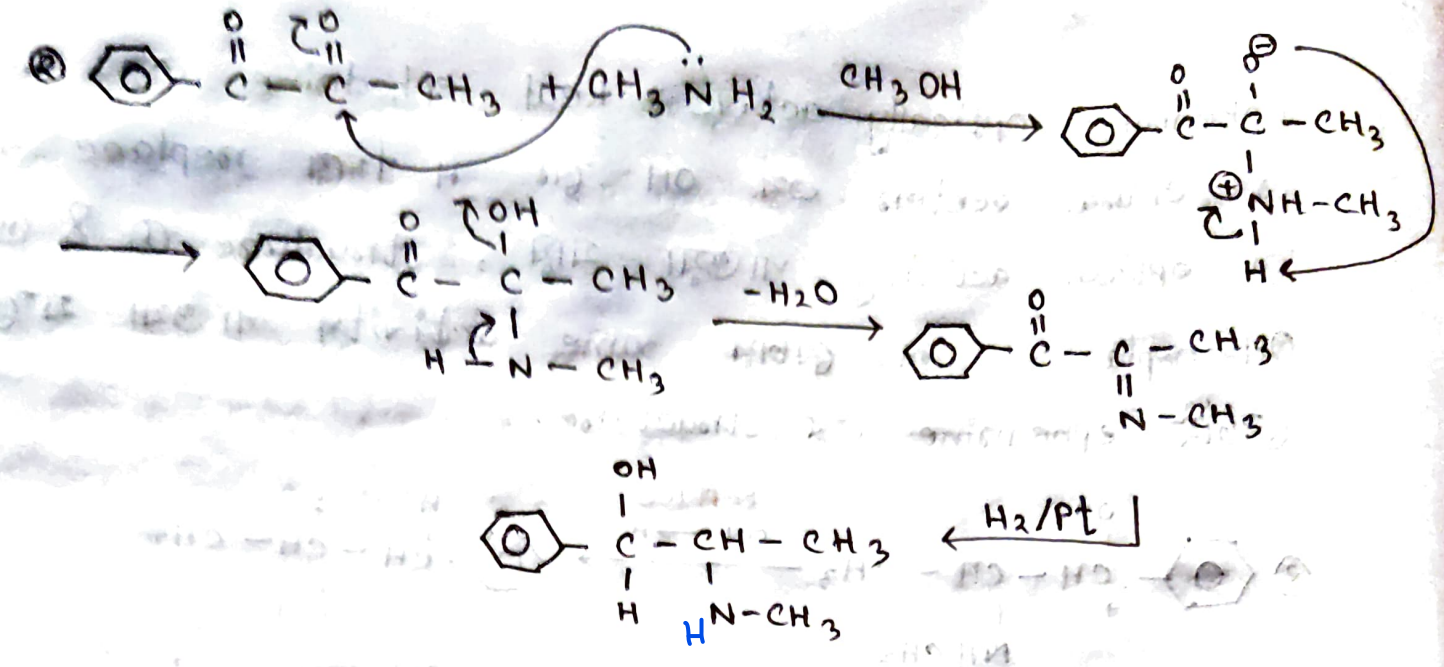
* ঔষধ molecule কখন chi optically active?

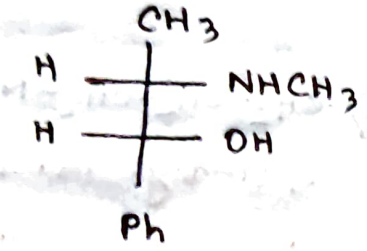
— chiral center
— chiral axis
— chiral plane

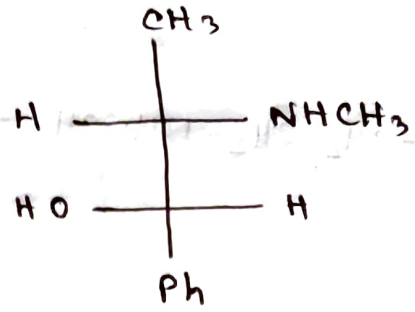
} এগুলো থাকবে

* Synthesis of Ephedrine : (Lab এ)



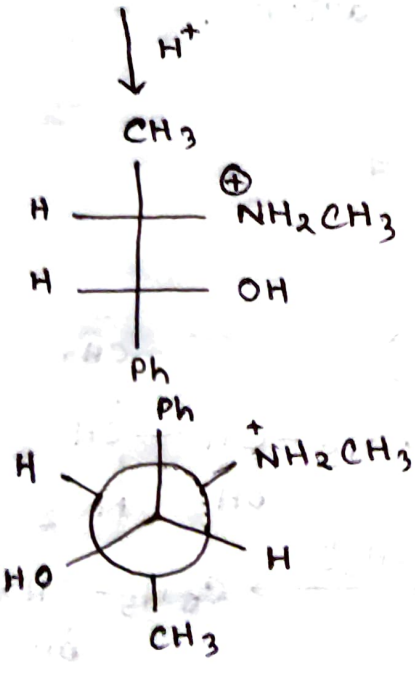


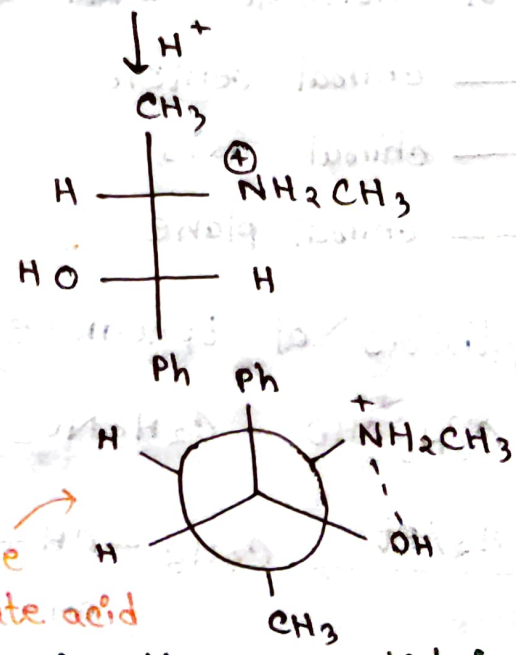




D (-) Ephidrin

L (+) Ephidrine





forms stable conjugate acid

* Why L-ephedrine is more basic than D-ephidrine?
L-ephedrine forms more stable conjugate acid than D.
अर्थात् L-ephedrine more basic.

Atropine : এটা চোখের pupil এর কোন defect এ use হয়।

Molecular formula ⟶ $C_{17}H_{23}NO_3$

Double bond equivalent :

$$C_{17}H_{36+1}$$
$$- \quad C_{17}H_{37}$$
$$\overline{C_{17}H_{23}}$$

$H = 14$ $\quad \therefore DBEV = \frac{14}{2} = 7$

Melting Point : 118°C

Atropine as an esten —

$$C_{17}H_{23}NO_3 + H_2O \xrightarrow{Ba(OH)_2} C_9H_{10}O_3 + C_8H_{15}NO$$

(esten) — tropic acid (−COOH) — tropine (−OH)

Atropine is named as tropinetropate.

Tropic Acid :

① Molecular Formula = $C_9H_{10}O_3$

② DBEV = 5 (এটা benzene ring থাকবে double bond / cycle থাকতে পারে)

③ Tropic acid consumes one equivalent of alkali & does not add on Br, i.e., it is a saturated mono-basic acid.

④ $C_9H_{10}O_3 \xrightarrow[-H_2O]{\Delta} C_9H_8O_2 \xrightarrow{[O]}$ 

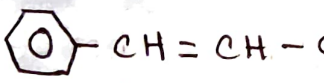
Tropic acid when heated strongly, it loses a molecule of H_2O to yield an unsaturated acid. Further oxidization, gives benzoic acid.

[যেহেতু ১ mole H_2O বেরিয়ে যায়, বলা যায় এটা -OH group আছে। আবার benzoic acid produce হওয়ায় বলা যায়, mono-substituted benzene ring আছে]

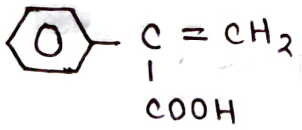
Tropic acid contains one benzene ring with one side chain.

(v) Tropic acid when acetylated forms a mono-acetylated derivative, indicating one -OH is present.

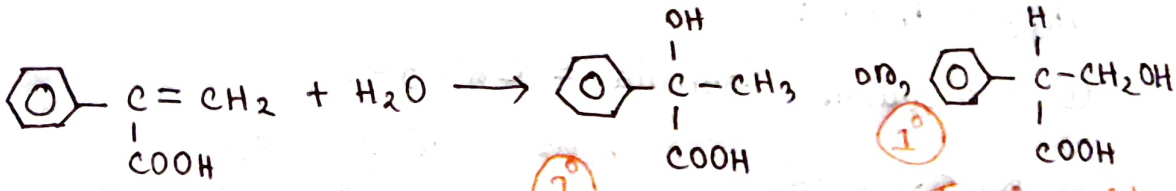
Possible structures of Atropic Acid :



→ But it is a known structure of Cinnamic Acid

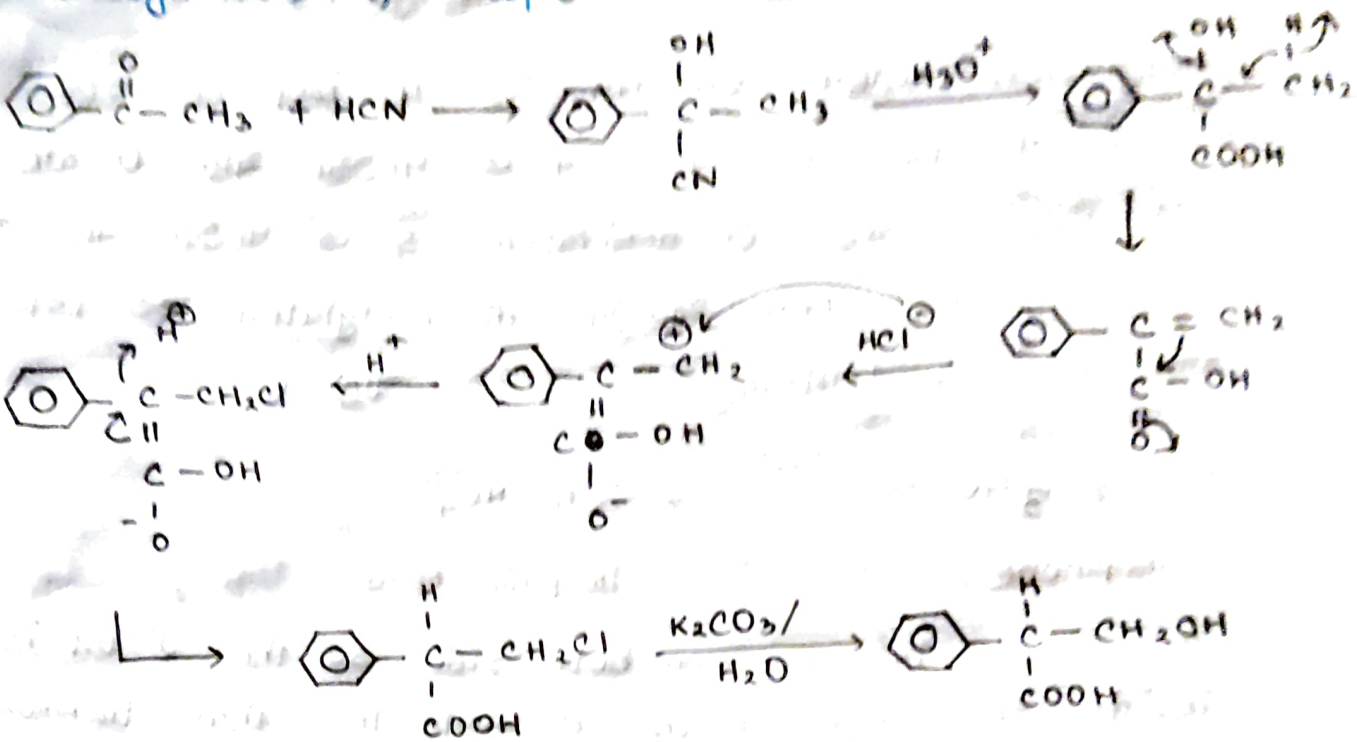


→ Hence, It is atropic acid

(vi) 

∵ 1 mole পানি বেরিয়ে যাওয়ায় Atropic acid হয়, তাই- 1 mole H_2O add করলে tropic acid পাওয়া যায়, আবার (iv) step -এ 1 বার oxidization -এ  পাওয়া যায় তাই বলা যায় alcohol 1° ছিল। তাই (vi) নং এ 1° product - ই tropic acid.

Lab Synthesis of Tropic Acid :



Structure of Tropine :

1. Molecular Formula : $C_8H_{15}NO$

2. DBEV = 2 [মানে benzene ring নাই , double bond ২টা বা ১টা triple bond বা cycle থাকতে পারে]

3. $C_8H_{15}NO \xrightarrow[CCl_4]{Br_2}$ No rxn [unsaturation test]

মানে double / triple bond নাই ।

4. $C_8H_{15}NO \xrightarrow{CrO_3} C_8H_{13}NO \longrightarrow$ ketone (Tropinone)

যেহেতু ketone produce হয়, বলা যায় 2° alc is present.

5. $C_8H_{15}NO \xrightarrow{Ac_2O}$ Mono-acetylated derivative

Acetylation করলে OH/NH এর H replace হয়ে acetylated product হয়। 1 এ পাওয়া যায় O alc রূপে আছে। মানে যে derivative 5 এ পাওয়া যায়, এ OH এর জন্য 1-NH থাকলে di-acetylated হত। তাই বলা যায় N tertiary form এ আছে।

6. $C_8H_{13}NO \xrightarrow{[O]} C_8H_{13}NO_4$

Tropinone — Tropinic Acid

C-number is same on both side. So the 2° OH is inside the ring. If the C would less than tropinone then the OH would be outside the ring.



outside ring এ থাকলে oxidation করলে bond cleavage-এ C-সংখ্যা কমে যেত, যেহেতু কমে নাই তাই বলা যায় alc ring এর ভিতরেই ছিল।



21.04.2024

⑦ $C_8H_{15}NO \xrightarrow[<150^\circ C]{HI} C_8H_{14}NI \xrightarrow[HCl]{Zn} C_8H_{15}N \xrightarrow{HCl} C_7H_{13}N + \boxed{CH_3Cl}$

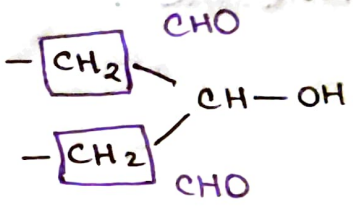
Tropine

Tropine is an alcohol

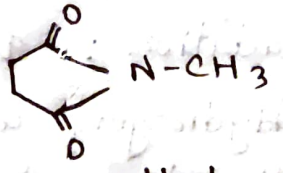
এই rxn দিয়ে N এর সাথে কয়টা methyl group আছে তা বের করা হয়। যত mole CH_3Cl হবে, ততটা methyl থাকবে।

Since one CH_3Cl is produced, N- is attached with one $-CH_3$ group.

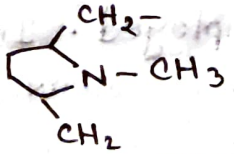
⑧ When 2 mole benzaldihide is added to tropine, ~~dibenzald~~ dibenzilidine derivative is produced. So two methylene group ($-CH_2$) is present in the vicinity of 2° OH group.



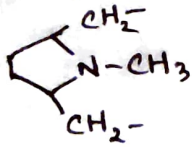
২টা $-CH_2$ free থাকলেই শুধু তারা b dibenzilidine compound তৈরি করবে।

⑨ $C_8H_{13}NO \xrightarrow[H_2SO_4]{CrO_3}$ 

N-methyl succinamide

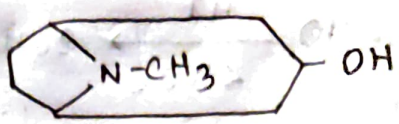
Tropine এর মধ্যে 

এরূ মত ring থাকলে তাকে strongly oxidize করলে N-methyl succinamide পাওয়া যায়।

So, 

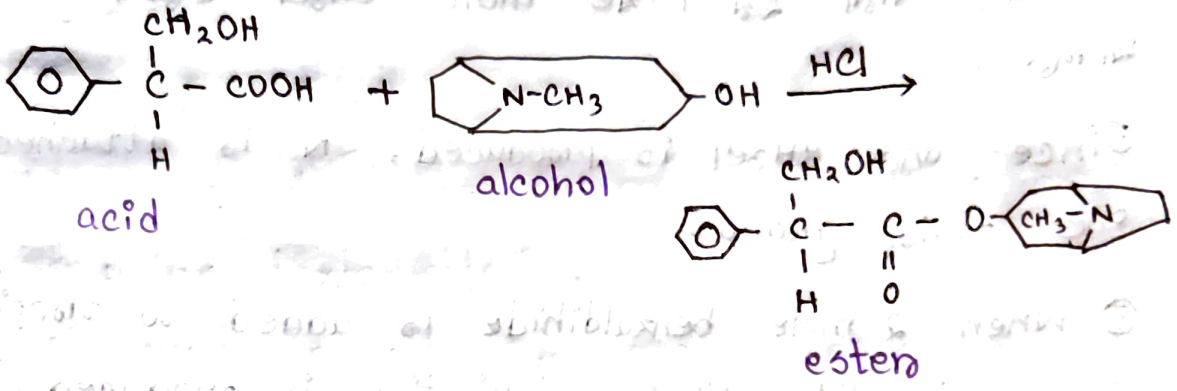
is present in tropine structure.

Structure of Tropine :

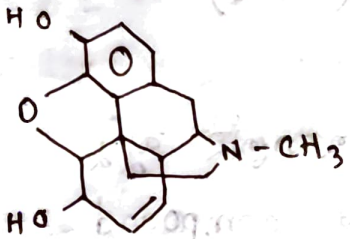


Atropine :

Atropine ⟶ Tropic Acid + Tropine



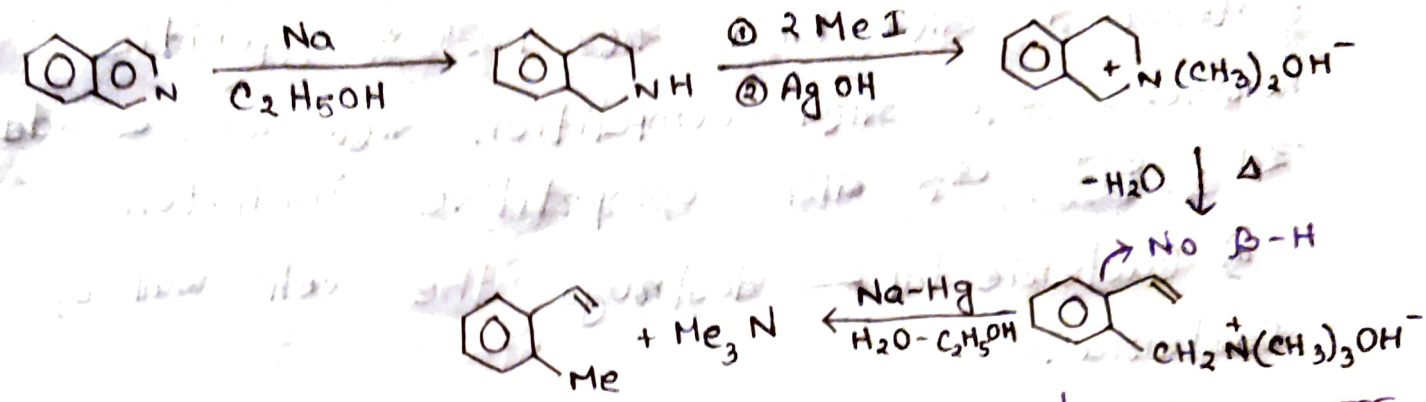
* Morphine :



* Emde's Degradation : If the alkaloid does not contain a β-Hydrogen atom, the Hoffmann's exhaustive methylation fails. In such case, Emde's method may be employed. In this method, the final step involves reductive cleavage of quarternary ammonium salts either with sodium in liquid NH_3 OR by catalytic hydrogenation.

Ex :

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↳ এতে β-H না থাকায় Hoffmann করা যায় না, emde's method করা হয়